

STUDENT CASE STUDY 1: PROBLEM STATEMENTS & PLANNING

Course: CMU-CS 445 – *System Integration Practices*

Semester: 2

Academic: 2024–2025

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Objective:

This case study helps students:

- Identify current business issues and integration requirements within an enterprise context.
- Analyze data inconsistency and operational inefficiency between multiple systems.
- Establish a foundation for the system integration project that will be developed through Case Studies 2 – 5.

Project Scenario

Company X operates two independent systems:

System	Database	Primary Purpose
HUMAN	SQL Server	Manages employee, department, and position information, ...
PAYROLL	MySQL	Handles salary, attendance, and tax records, ...

Currently, these systems are **not integrated**, which results in:

- Time-consuming manual aggregation of HR and payroll data.
- Inconsistent or duplicated employee information.
- Delays in financial reporting and profit calculations.
- Incomplete management reports for decision-making.

The **CEO of Company X** has issued a memo requesting a new integrated solution that connects both databases and delivers a unified dashboard for HR, payroll, and shareholder analytics.

Tasks and Requirements

Each student group must produce **three main design documents** and define the **project scope** as described below.

1. Project Proposal Document

Purpose: Establish a clear understanding of the business problem and propose a feasible system integration solution.

Content Requirements:

- **Business Problem Analysis:**

- Analyze issues described in the CEO Memo.
- Explain how existing HR and Payroll systems cause operational inefficiencies, such as:
 - Time-consuming data aggregation.
 - Inconsistent records between databases.
 - Lack of integrated reporting capability.

- **Proposed Solution:**

- Describe the integration approach to be applied (e.g., API-based, ETL, or Hybrid).
- Specify technologies, frameworks, and tools planned for implementation (e.g., Python Flask, ASP.NET MVC, RESTful API, SQL Server Connector).

- **Project Plan Overview:**

- Outline milestones and an estimated implementation timeline.
- Provide an initial cost or resource estimation if applicable.

2. Project Plan Document

Purpose: Present a structured roadmap for project execution and teamwork management.

Content Requirements:

- **Project Roadmap:**

- Define development phases aligned with the following Case Studies (CS2 – Integration Models, CS3 – API, CS4 – Security, CS5 – Dashboard).

- **Detailed Timeline:**

- Assign weekly tasks and expected outputs for each team member.

- **Roles and Responsibilities:**

- Assign specific duties: Project Manager, Data Integrator, API Developer, Security Engineer, Dashboard Designer.

3. Business Requirements Document (BRD)

Purpose: Define the functional and non-functional requirements for the integrated system.

Business Requirements:

- **Data Integration:**
 - Enable synchronization of employee and payroll data between *HUMAN* and *PAYROLL* databases.
 - Ensure automatic updates for employee information, salaries, and department changes.
- **Overview Dashboard:**
 - Display real-time KPIs including:
 - Total Employees (by department / employment status).
 - Total Payroll Fund (monthly salary expenditure).
 - Shareholder Profit (total profit and distribution summary).
- **Reporting System:**
 - Generate structured HR and Payroll reports:
 - Employee count by department.
 - Salary reports (monthly salaries and breakdowns).
 - Shareholder profit reports (quarterly and annual profit allocation).
- **Smart Alert System:**
 - Work Anniversary Alerts: Display employees reaching 1-, 5-, and 10-year milestones.
 - Leave Alerts: Notify managers when an employee exceeds the annual leave limit.

4. Project Scope

To maintain system stability and ensure realistic implementation:

- **Database Structure Limitation:** No modifications are allowed to the schema of *HUMAN_2025* or *PAYROLL*.
- **Independent Development:** The new dashboard must function as a separate application that connects to both databases through integration APIs or middleware.
- **Focus Areas:** Data consistency, performance, security, and reporting accuracy.

5. Expected Deliverables

5.1 Document Deliverables

Document Name	Action (New / Update)	Detailed Description / Student Instructions
1. Project Proposal Document	New	Write a professional proposal based on the CEO Memo. Clearly identify current issues between the HR (HUMAN) and Payroll (PAYROLL) systems such as duplicated data, slow reporting, or lack of synchronization. Propose an integration solution, list the technologies to be used (e.g., API, SQL Server, MySQL, React/ASP.NET MVC), and include estimated timeline, milestones, and cost.
2. Project Plan Document	New	Create a simple project plan showing the timeline (Gantt chart or table) and member responsibilities. Divide tasks by Case Study stages (Case 1–5). Include who is in charge of documentation, backend, frontend, testing, and presentation.
3. Business Requirement Document (BRD)	New	Document all business requirements collected from the CEO Memo. Describe what the company needs (e.g., integrated employee and payroll data, consistent reports, automatic alerts). For each requirement, explain its business value. Example: “Integration between HR and Payroll ensures salary data consistency and reduces manual entry errors.”

6.2 Software Functional Deliverables

Case Study	Software Functions to Develop	Description / Implementation Guidance
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CASE STUDY 1 – Problem Identification & Proposal (Week 1–2)	<i>No coding required</i>	In this Case study, students analyze problems and design the foundation for the integration project. Deliverables are analytical documents only (Proposal, BRD, Plan). However, students must clearly define the core functions the system will later implement: • Employee management • Payroll management • Attendance tracking • Reports and Alerts These functions will be developed step by step in later Case Studies.
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6. Evaluation Criteria

Criteria	Description	Weight
Problem Analysis	Clear understanding of business issues and requirements	25%
Solution Feasibility	Appropriateness of integration model and tools proposed	25%
Documentation Quality	Completeness and professionalism of deliverables	25%
Teamwork & Presentation	Collaboration and clarity in presenting findings	25%
Total		100%